

Week 15: Bonus: monad transformers

Functional Programming in Lean

Learning goals

- ▶ `StateT` and `ExceptT`
- ▶ Effect-stack result types
- ▶ State after an error

Core concepts

- ▶ `StateT Nat (Except String) A`: error or value with state
- ▶ `ExceptT String (StateM Nat) A`: error or value, plus state
- ▶ Transformer order and failure semantics

Lean example

```
abbrev LoseState :=  
  StateT Nat (Except String)  
abbrev KeepState :=  
  ExceptT String (StateM Nat)
```

Lean details

- ▶ `failLose`: discarded final state
- ▶ `failKeep`: final state 1
- ▶ Result type as the explanation

Live coding

- ▶ State increment before error
- ▶ Evaluation of both stacks
- ▶ Comparison of outputs

Check

Stack retaining state after failure

Exercise

- ▶ Expansion of both result types
- ▶ Error trace